

INFORMATION TECHNOLOGY INSTITUTE

LAB 6

Differential Amplifier





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Part 1: Differential Amplifier Design

Sizing

Input pair sizing

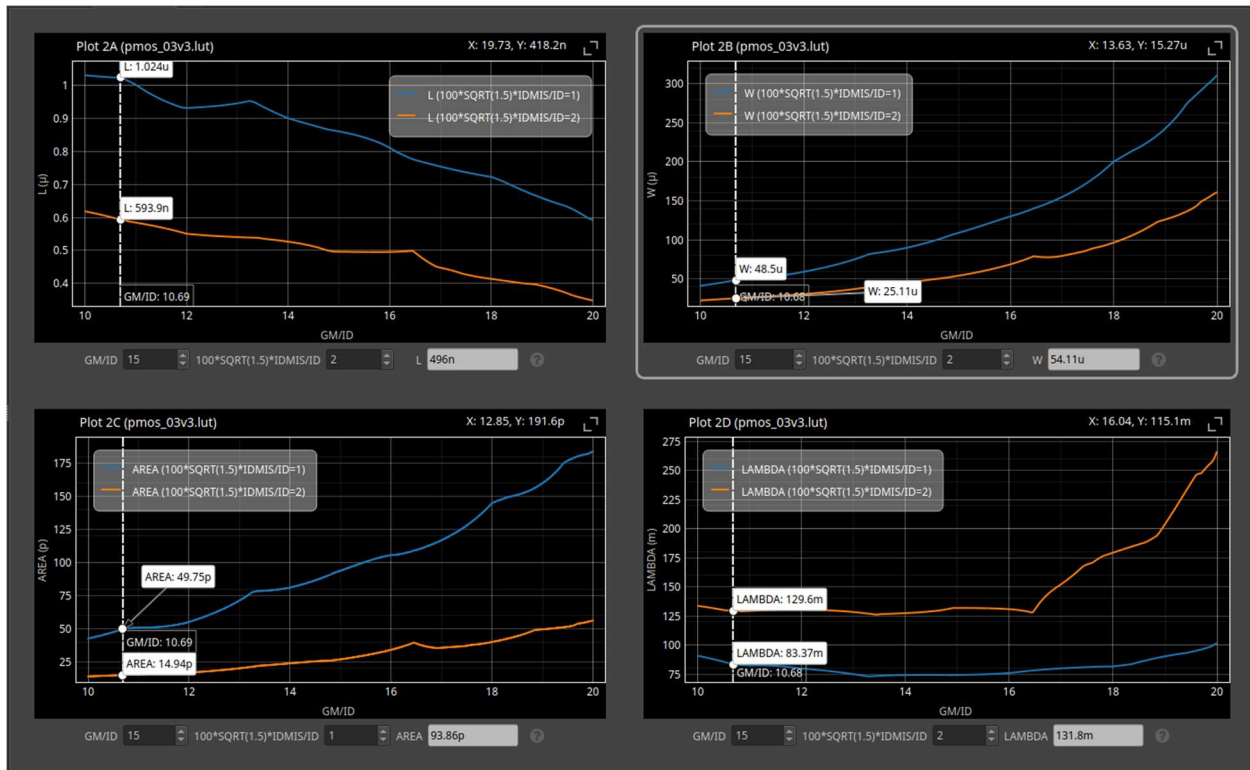
- We want an output Common mode of 0.6v and we set a V_{ds} for the the simple current mirror of 0.3v so the V_{ds} for our input pair is 0.9v
- We have an ISS current of 40uA it will be divided across each half so each mosfet will carry 20uA
- We want $r_o \gg R_d$, $r_o = 10R_d$ so $r_o = 10 \cdot 0.6 / I_D$
- We want a DC diff gain of 8 and $A_{vd} = g_m(r_o \parallel R_D) = 0.91 g_m R_D = 0.91 \left(\frac{g_m}{I_D} \right) 0.6$ so $\frac{g_m}{I_D} = \frac{8}{0.6 \times 0.91} = 14.65 \text{ V}^{-1}$
- $R_D = 0.6 / 20\mu = 30\text{k}\Omega$
- Lastly $V_{sb} = 0.3\text{v}$

Parameter	
Supply (V_{DD})	1.8V
Current seed (Ref current)	20 μ A
Bias current (I_{SS})	40 μ A
Differential gain	8
CM output level ¹	$V_{DD}/3$
Load capacitance	1pF

Plot 1A		Plot 1B	Plot 1C	Plot 1D
Import:		Plot 1B	OK	?
LUT Settings				
LUT		pmos_03v3	?	
Corner		TT	All	?
Temp (°C)		27.0	All	?
Frequency		1	?	
ID		20u	?	
gm/ID		8/(0.6*0.91)	?	
ro		10*(0.6/ID)	?	
VDS		0.9	?	
VSB		0.3	?	
Stack		1	?	
Results:				
	Name	TT-27.0		
1	ID	20u		
2	IG	N/A		
3	L	350n		
4	W	30u		
5	VDB	1.2		
6	VGB	1.241		
7	VGS	941.1m		
7	VGS	941.1m		
8	VDS	900m		
9	VSB	300m		
10	VGB	1.241		
11	VDB	1.2		
12	gm/ID	14.24		
13	Vstar	140.5m		



Current mirror sizing



Parameter	
Input current	20μA
Percent mismatch: $\sigma(I_{out})/I_{out}$	≤ 2%
Lambda	0.1
Compliance voltage	≤ 200mV
Area	Minimize

We choose a point where our mosfet is in the strong inversion so we can minimize area and it already have fulfilled the rest of design design parameters

$Gm/I_d=10.69$, $v^*=187mV$



Calculation of CM input level

$$V_{DG} < V_{TH}$$

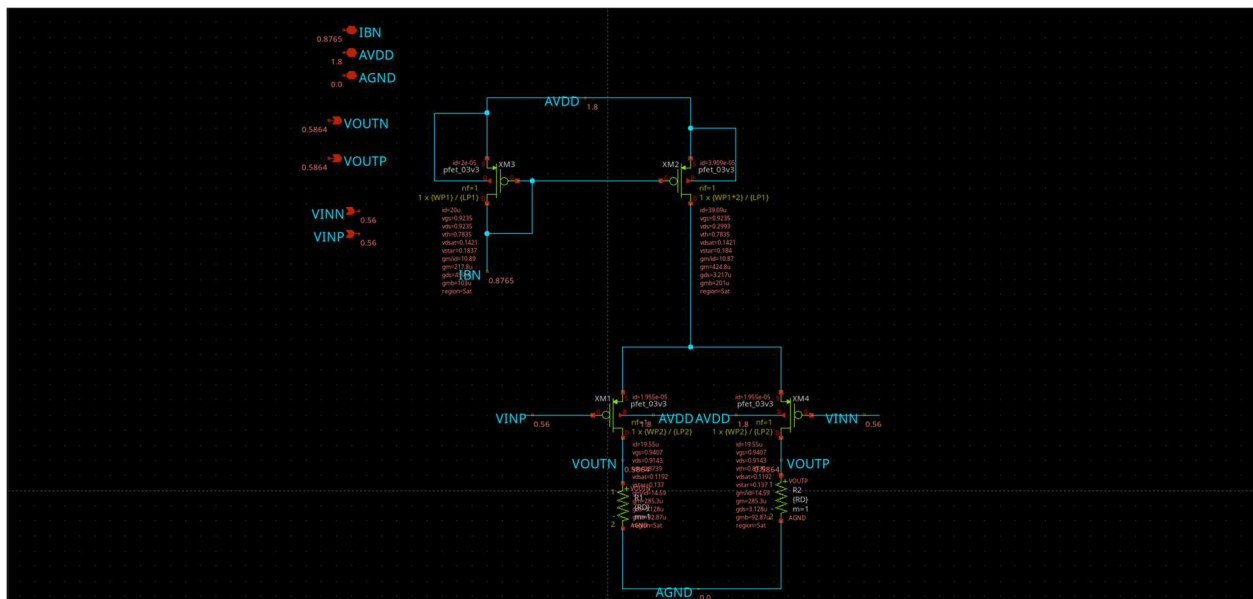
$$0.6 - V_{ICM} < 0.87 \Rightarrow V_{ICM} > -0.27 \text{ V}$$

$$V_{ICM} < V_{OV} + V_{TH} + V_{OV}$$

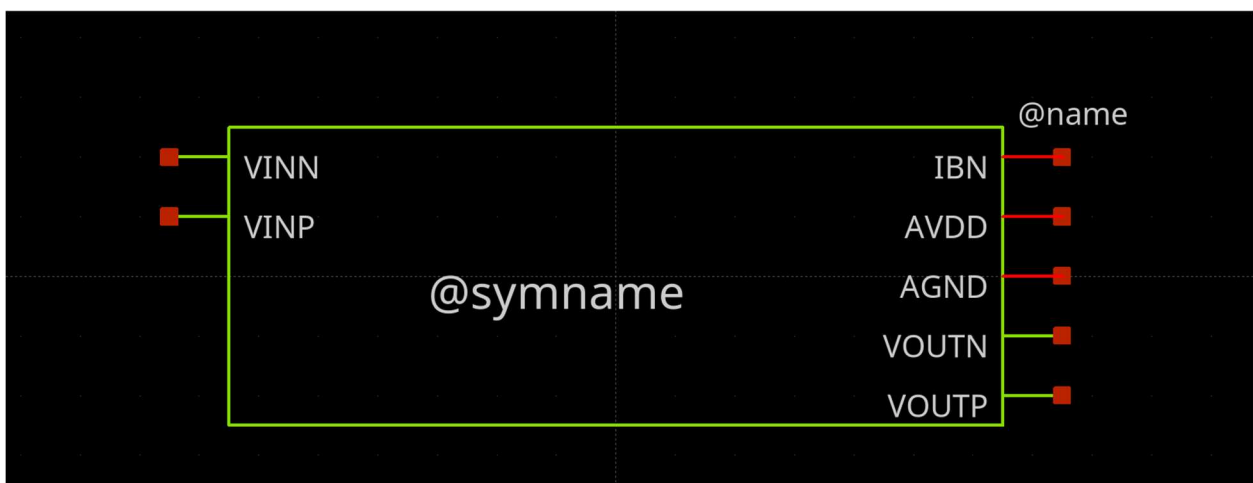
$$V_{ICM} < 0.140 + 0.87 + 0.187 \Rightarrow V_{ICM} < 1.197 \text{ V}$$

So yes the input used is $1.8 - 0.3 - 0.94 = 0.56 \text{ V}$ which is absolutely suitable

Symbol schematic



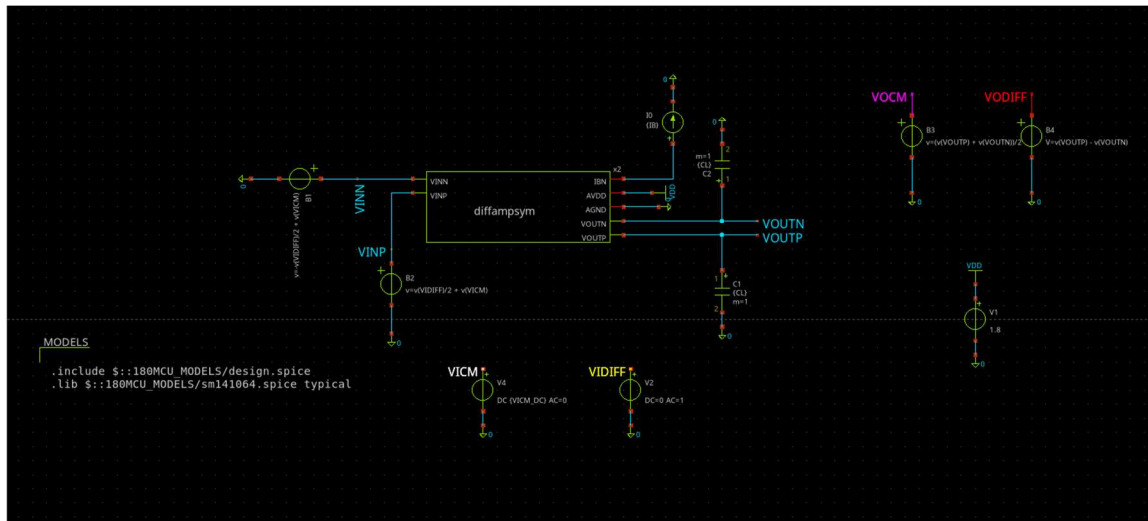
Symbol



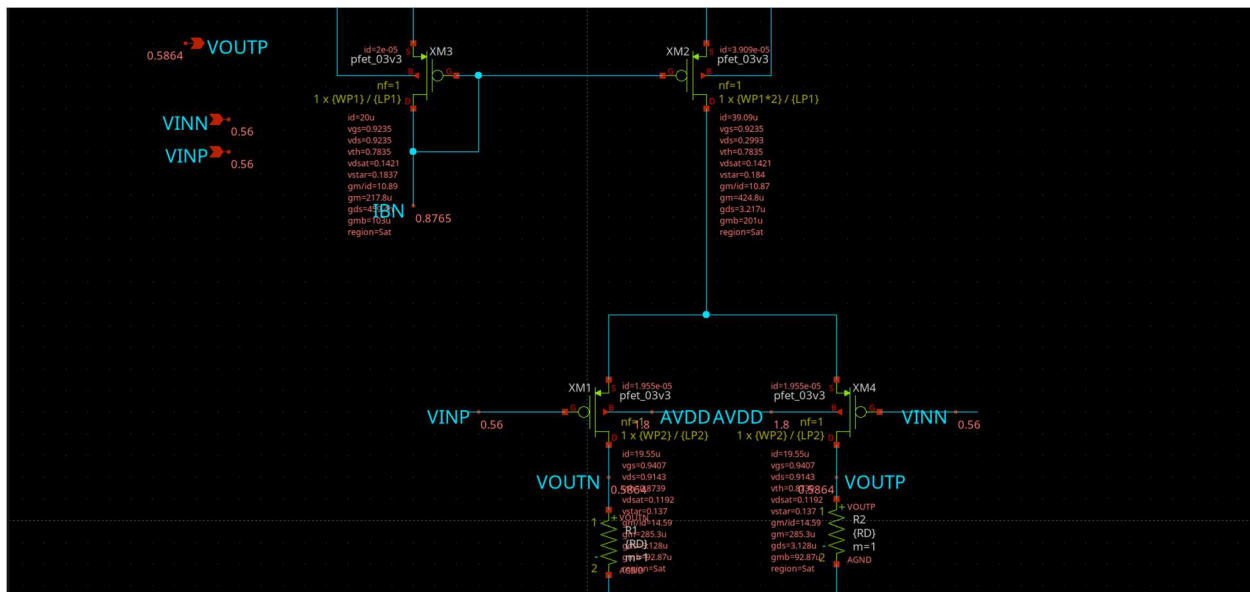


Part 2: Differential Amplifier Simulation

Schematic



Annotated OP

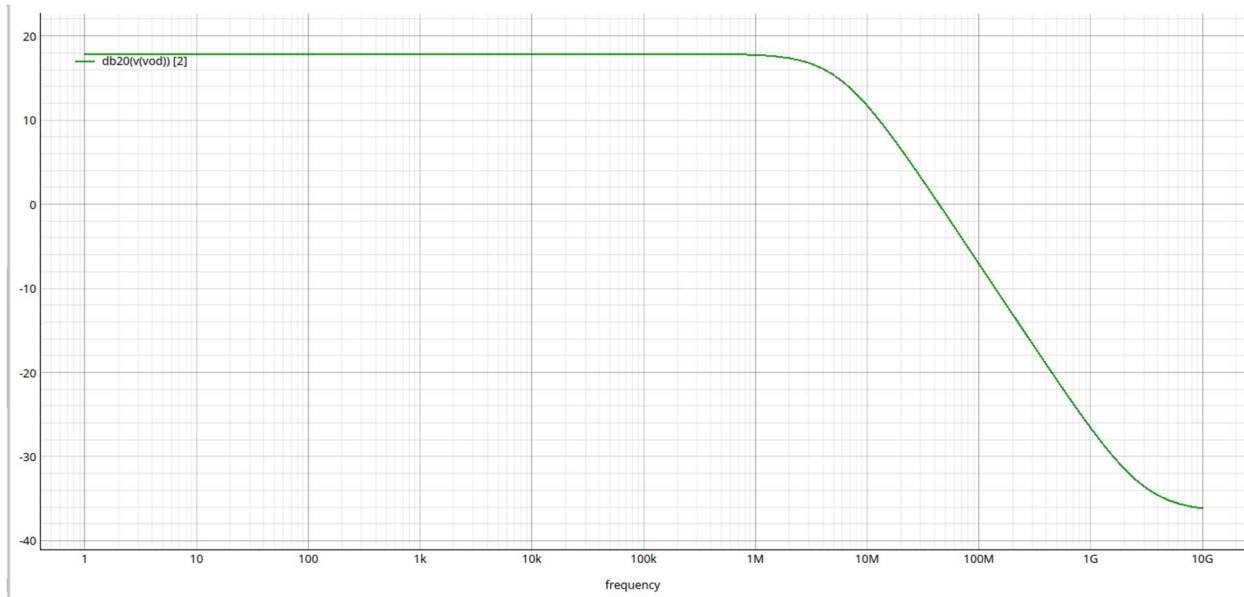


All transistors are in sat as $V_{DS} > V_{DSAT}$

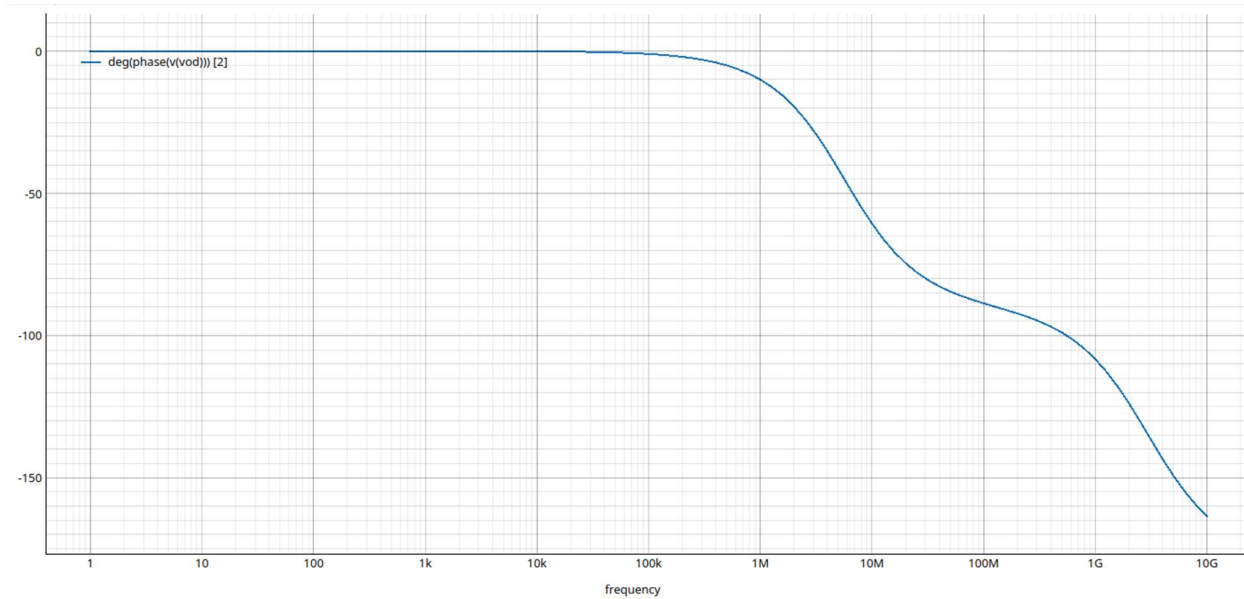


Diff small signal ccs

Bode Plot



Phase





Comparison

POC	Simulated	Hand analysis
Gain	7.824=17.86dB	7.82 = 17.86 dB
BW	5.68MHz	5.80MHz

hand analysis

$$r_o = \frac{1}{g_{ds}} = \frac{1}{3.128 \times 10^{-6}} = 319.7 \text{ k}\Omega, R_{out} = r_o \parallel R_D = 319.7 \parallel 30 = 27.43 \text{ k}\Omega$$

$$A_{vd} = g_m R_{out} = (285 \times 10^{-6})(27.43 \times 10^3) = 7.82 \text{ V/V}$$

$$A_{vd,dB} = 20 \log_{10}(7.82) = 17.86 \text{ dB}$$

$$C_L = 1 \text{ pF}, \quad R_{out} = 27.43 \text{ k}\Omega$$

$$BW = \frac{1}{2\pi R_{out} C_L} \quad BW = \frac{1}{2\pi(27.43 \times 10^3)(1 \times 10^{-11})} = 5.80 \times 10^6 \text{ Hz} \quad \boxed{BW \approx 5.80 \text{ MHz}}$$

```
=== Bias Points ===
M1: gm/id = 14.5947 S/A, Vstar = 0.137036 V
M2: gm/id = 10.8658 S/A, Vstar = 0.184063 V
M3: gm/id = 10.8882 S/A, Vstar = 0.183686 V
M4: gm/id = 14.5947 S/A, Vstar = 0.137036 V
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

No. of Data Rows : 101
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

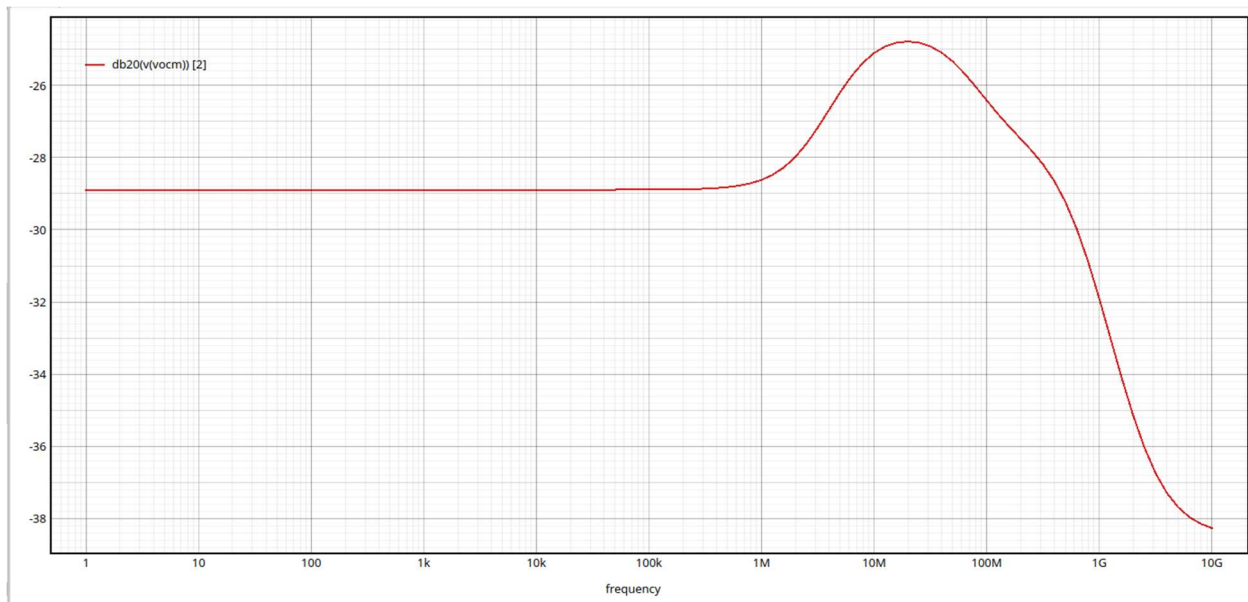
Using SPARSE 1.3 as Direct Linear Solver

No. of Data Rows : 101
av_diff_db      = 1.786882e+01
av_diff_mag     = 7.824223e+00
bw_diff         = 5.684065e+06
av_cm_db        = -2.888829e+01
av_cm_mag       = 3.594061e-02
bw_cm           = 9.980193e+08
=== AC Results ===
Av Diff: 17.8688 dB, BW: 5.68406E+06 Hz
Av CM: -28.8883 dB, BW: 9.98019E+08 Hz
CMRR: 46.7571 dB
```

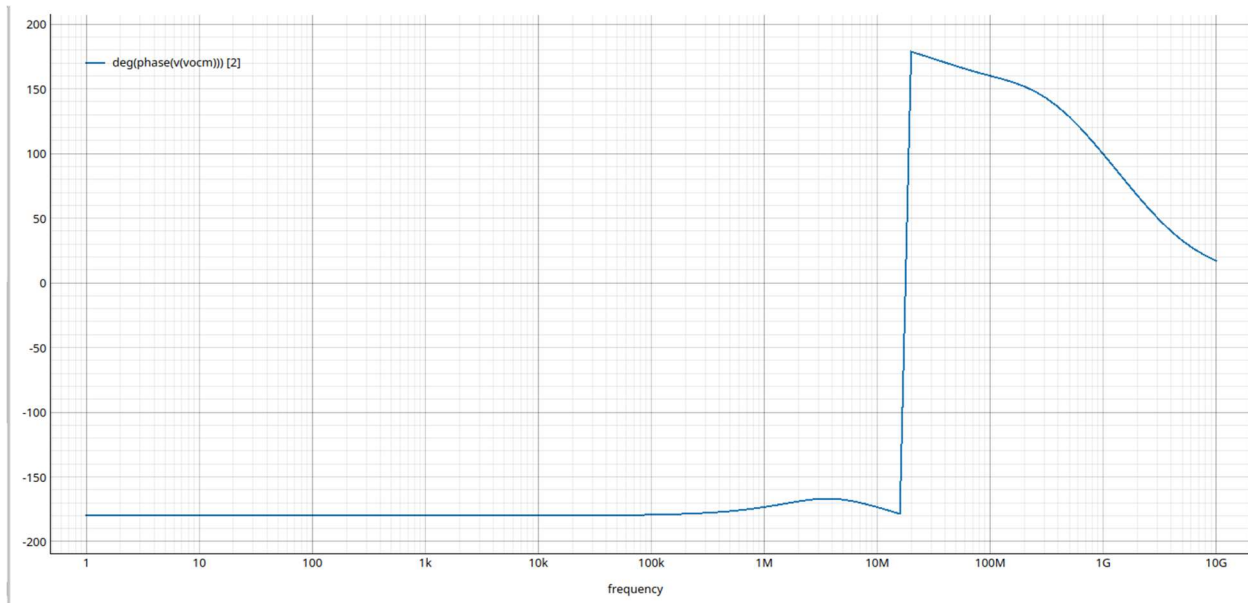


CM small signal ccs

Bode Plot



Phase



The phase here seems weird due to a plotting bug in Ngspice station because it should continue decreasing in negative but because of the limits of the program it rotated back to being positive but this completely normal as an angle +360 stays the same



Comparison

$$r_o = \frac{1}{g_{ds1}} = \frac{1}{3.128 \mu\text{S}} = 319.7 \text{ k}\Omega \quad R_{SS} = \frac{1}{g_{ds2}} = \frac{1}{3.217 \mu\text{S}} = 310.8 \text{ k}\Omega$$

$$R_L = R_D \parallel r_o = 30 \parallel 319.7 = 27.43 \text{ k}\Omega$$

$$A_{v,CM} = -\frac{g_m R_L}{1+2g_m R_{SS}}, A_{v,CM} \approx -\frac{g_m R_L}{2g_m R_{SS}} = -\frac{R_L}{2R_{SS}}$$

$$A_{v,CM} \approx -\frac{27.43}{2(310.8)} = -0.04412 \text{ V/V}$$

$$A_{v,CM,dB} = 20 \log_{10}(0.04412) = -27.11 \text{ dB}$$

Method	CM gain (V/V)	CM gain (dB)
Hand analysis	0.04412	-27.11 dB
simulation	0.03594	-28.89 dB

It is obviously less than one as the RSS is larger than $R_D \parallel r_o$ so the amplifier becomes an attenuator as the gain can be simplified to $R_D \parallel r_o / 2R_{SS}$

```

=== Bias Points ===
M1: gm/id = 14.5947 S/A, Vstar = 0.137036 V
M2: gm/id = 10.8658 S/A, Vstar = 0.184063 V
M3: gm/id = 10.8882 S/A, Vstar = 0.183686 V
M4: gm/id = 14.5947 S/A, Vstar = 0.137036 V
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

No. of Data Rows : 101
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

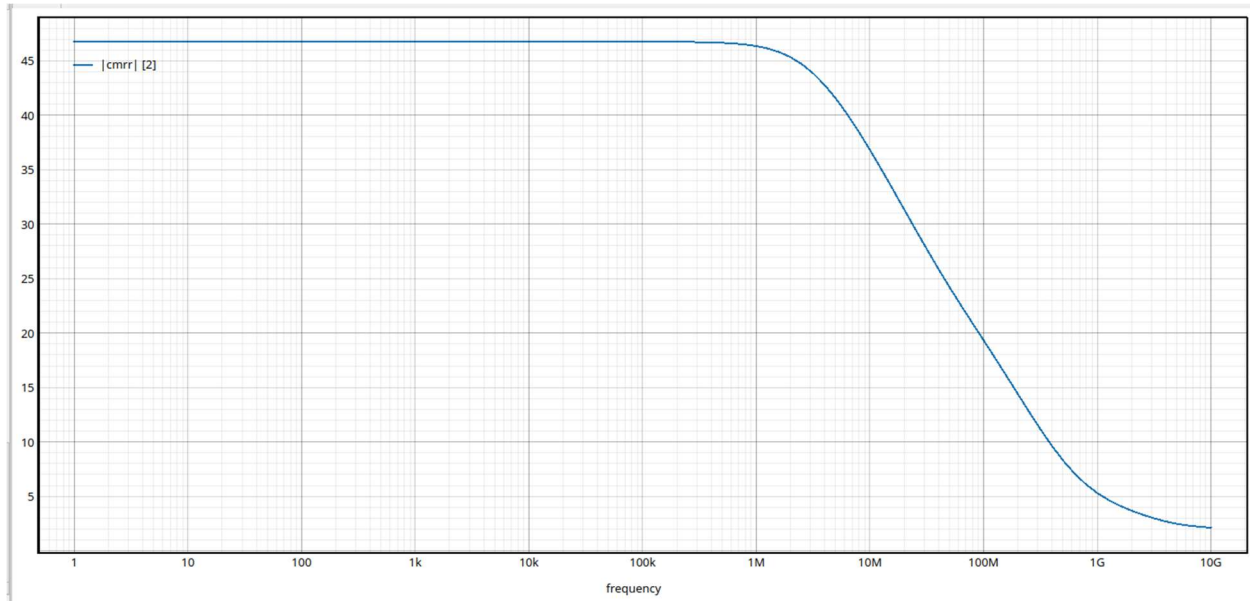
No. of Data Rows : 101
av_diff_db      = 1.786882e+01
av_diff_mag     = 7.824223e+00
bw_diff         = 5.684065e+06
av_cm_db        = -2.888829e+01
av_cm_mag       = 3.594061e-02
bw_cm           = 9.980193e+08
=== AC Results ===
Av Diff: 17.8688 dB, BW: 5.68406E+06 Hz
Av CM: -28.8883 dB, BW: 9.98019E+08 Hz
CMRR: 46.7571 dB

```



The reason behind the variation is due to the parasitic caps at the node of the tail Cs increasing Avcm and also due to the load capacitor as both nodes are associated with a pole which cause this change and the node is of the tail current is also associated with a zero that cause this increase in and this peaking effect.

CMRR Bode Plot



Comparison

$$\text{CMRR}(f) = \left| \frac{A_{vd}(f)}{A_{v,CM}(f)} \right| \quad \text{CMRR}_{\text{dB}}(f) = 20 \log_{10} \left| \frac{A_{vd}(f)}{A_{v,CM}(f)} \right|$$

$$\text{CMRR}_{\text{dB}} = A_{vd,\text{dB}} - A_{v,CM,\text{dB}}$$

$$\text{CMRR}_{\text{hand}} = \frac{7.82434}{0.0441169} = 177.35$$

$$\text{CMRR}_{\text{hand,dB}} = 20 \log_{10}(177.35) = 44.98 \text{ dB}$$

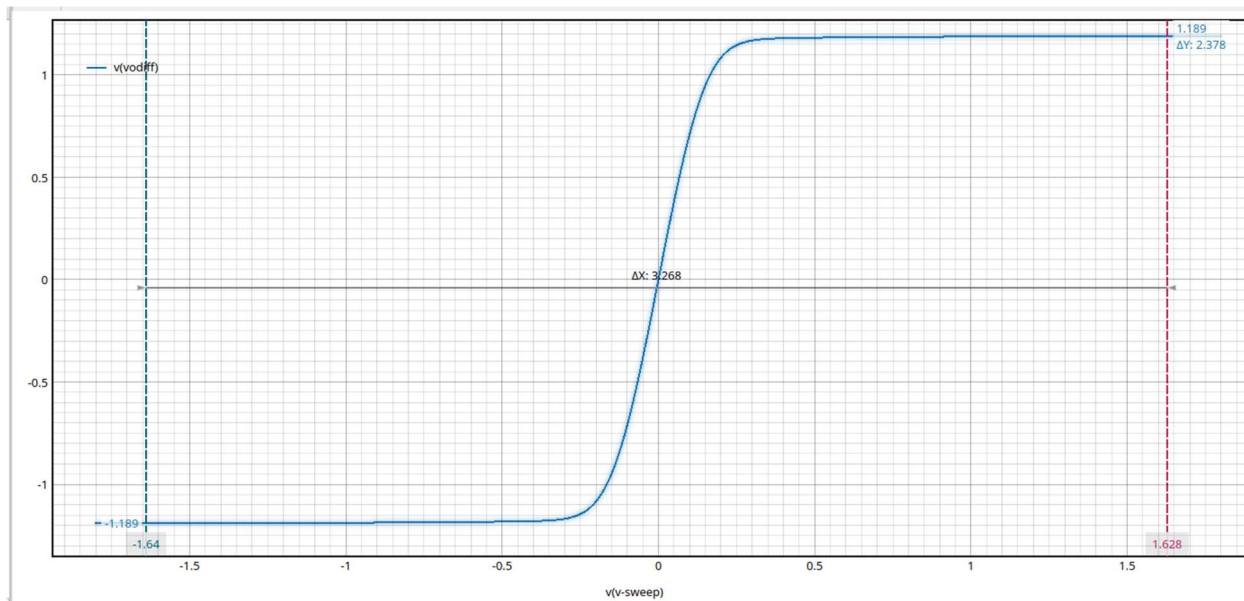
Method	$(A_{vd}/A_{v,CM})(V/V)$	$(A_{vd}/A_{v,CM})(\text{dB})$
Hand analysis	177.35	44.98 dB
simulation	217.70	46.76 dB



Justify the change due the frequency

As the tail node have R_{ss} it attenuates the CM signal heavily but as the frequency increases due parasitic capacitors the node impedance drops which increases the gain for the CM causing a worst CMRR

Diff large signal ccs



No. of Data Rows : 361
Diff Large Signal Analysis Results:
Max Output: 1.18892 V
Min Output: -1.18892 V

Comparison

$I_{tail} = 2I_B = 2(20 \mu A) = 40 \mu A$ for large positive V_{IDIFF} $I_4 \approx I_{tail}$, $I_1 \approx 0$

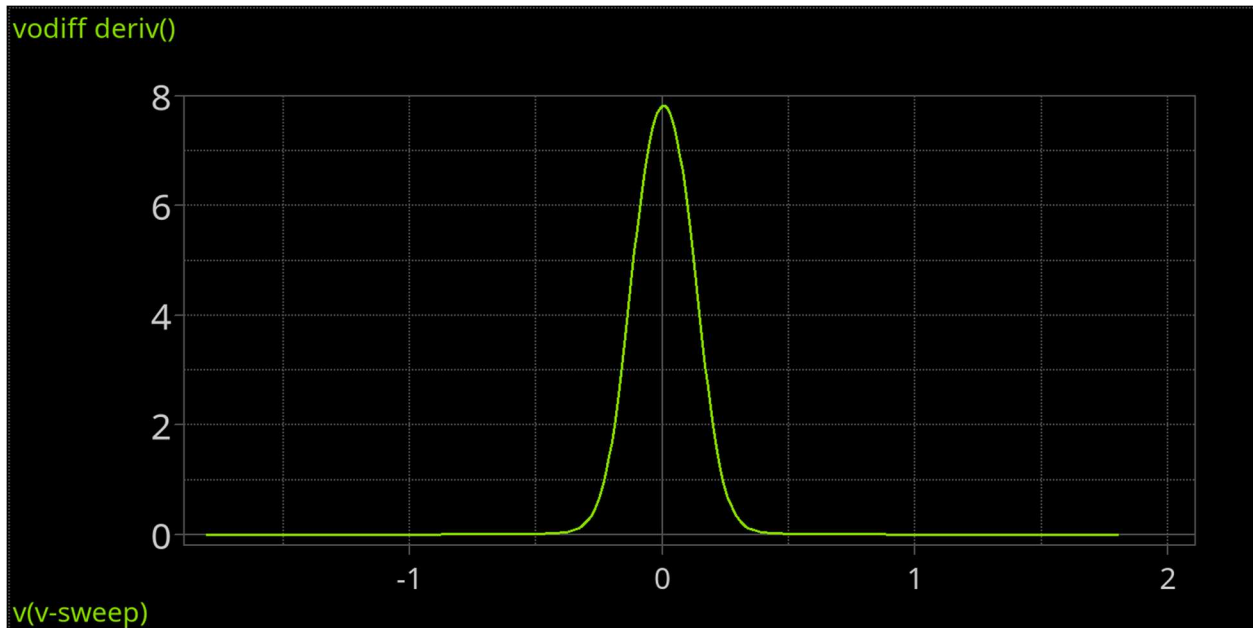
$V_{ODIFF,max} = I_{tail}R_D$, $V_{ODIFF,max} = (40 \times 10^{-6})(30 \times 10^3) = 1.20 V$

$V_{ODIFF,min} = -I_{tail}R_D = -1.20 V$

Method	Maximum(V_{ODIFF})	Minimum(V_{ODIFF})
Hand analysis	+1.200 V	-1.200 V
simulation	+1.18892 V	-1.18892 V



Plot derivative of the Vout

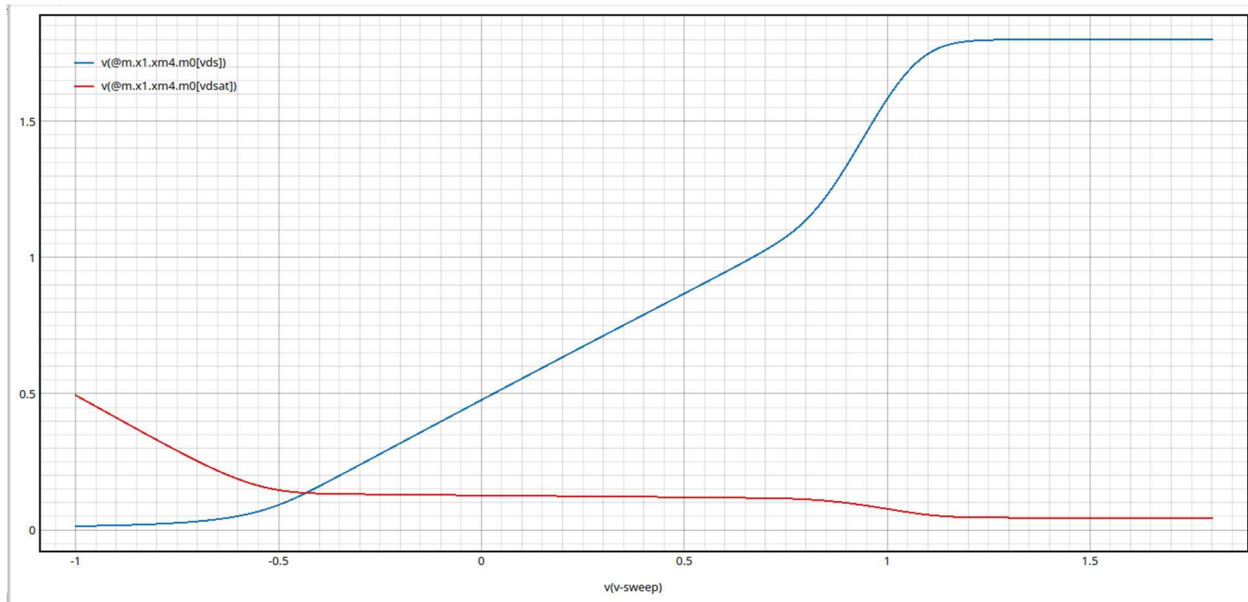


The peak value is around 7.9 which is exactly equal to our DC differential gain of 7.9 and it is because the derivative is $\Delta v_{out} / \Delta v_{id}$ which is equal to the gain and it is maximum at middle this is because it is when g_m is of both sides is the highest when the current is balanced in the two branched.

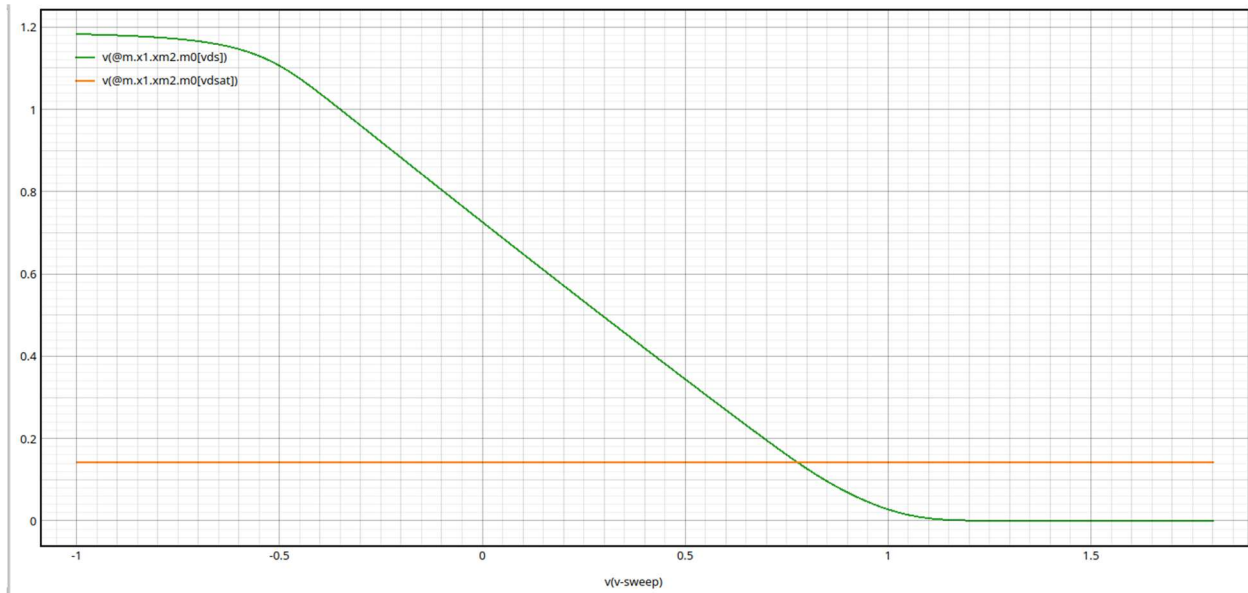


CM large signal ccs (region vs VICM)

Input pair

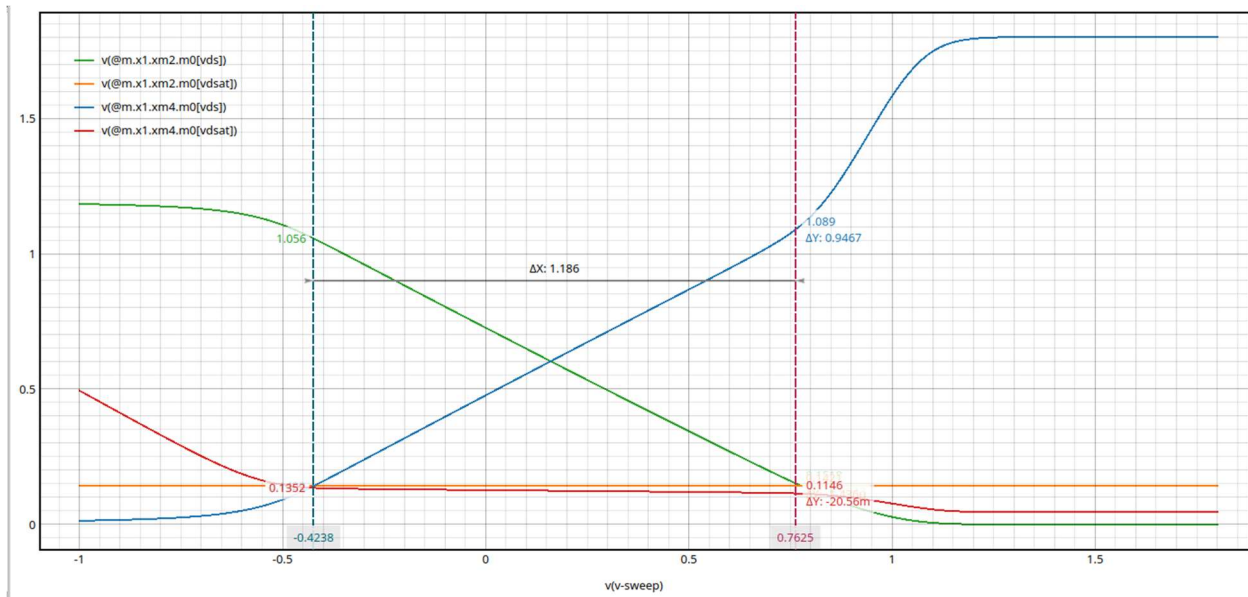


Tail source





overlaid



Comparison

$$V_{SD} \geq V_{SG} - |V_{TH}|, V_{ICM,min} = V_{OQ} - |V_{TH}|, V_{ICM,min} = 0.6 - 0.87 = -0.27 \text{ V}$$

$$V_{SD} \geq V_{OV2}, V_{ICM,max} = V_{DD} - |V_{TH}| - V_{OV} - V_{OV}$$

$$V_{ICM,max} = 1.8 - 0.87 - 0.140 - 0.187 = 0.603 \text{ V}$$

$$\boxed{-0.27 \text{ V} < V_{ICM} < 0.603 \text{ V}}$$

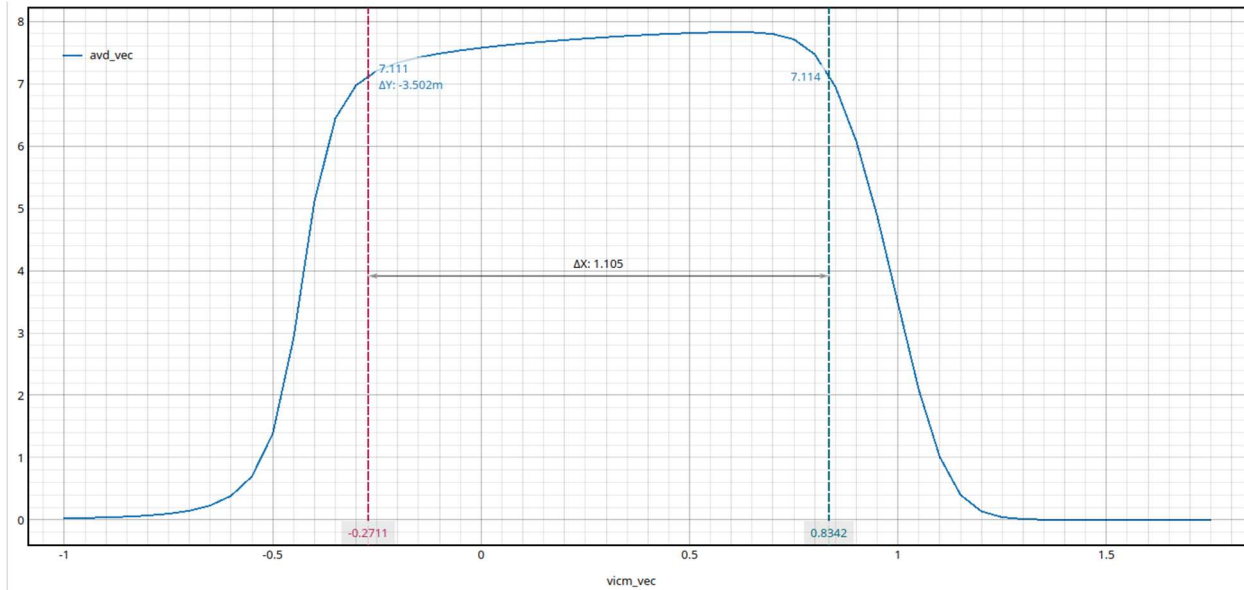
Method	Minimum(V_{ICM})	Maximum(V_{ICM})	CMIR width
Hand analysis	-0.270 V	0.603 V	0.873 V
simulation	-0.4238 V	0.7625 V	1.1863 V

Reason behind difference between analytical and simulation

Due the change in Change of V_{th} which is mainly because of body effect.



CM large signal ccs (GBW vs Vicm)



CM valid range is -0.2711v:0.8342v

Comparison

V_{Dsat} method	Avd vs V_{icm} method
-0.4238v : 0.7625v	-0.2711v : 0.8342v
1.1863v	1.1053v



Bonus

Sizing

PMOS I:V

calculating the VB range

$$VB < 1.8 - 0.187 - 0.050 - 0.187 - 0.75$$

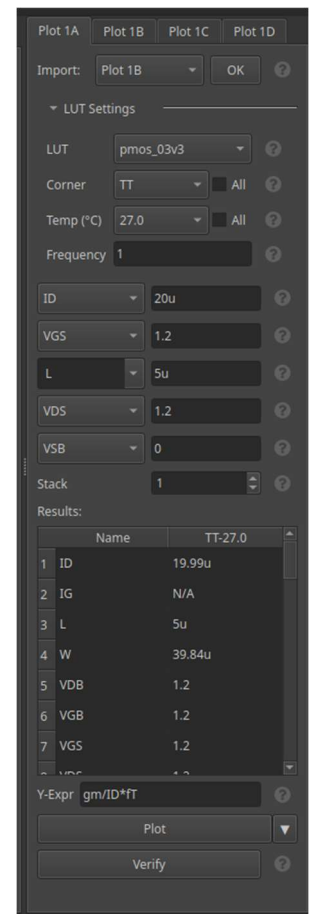
VBmax is 626mV

And minimum

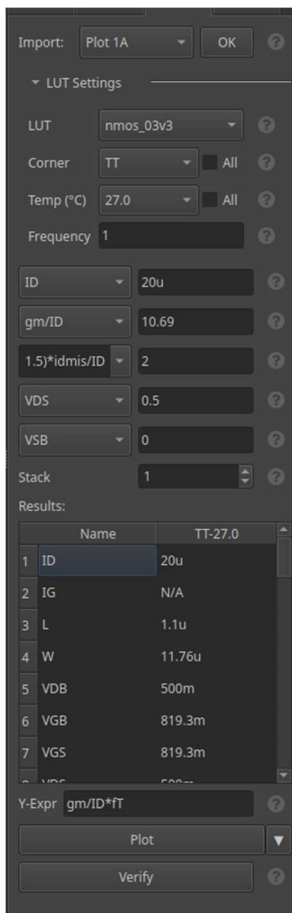
Is around 65mV

So we want it to be 600mV

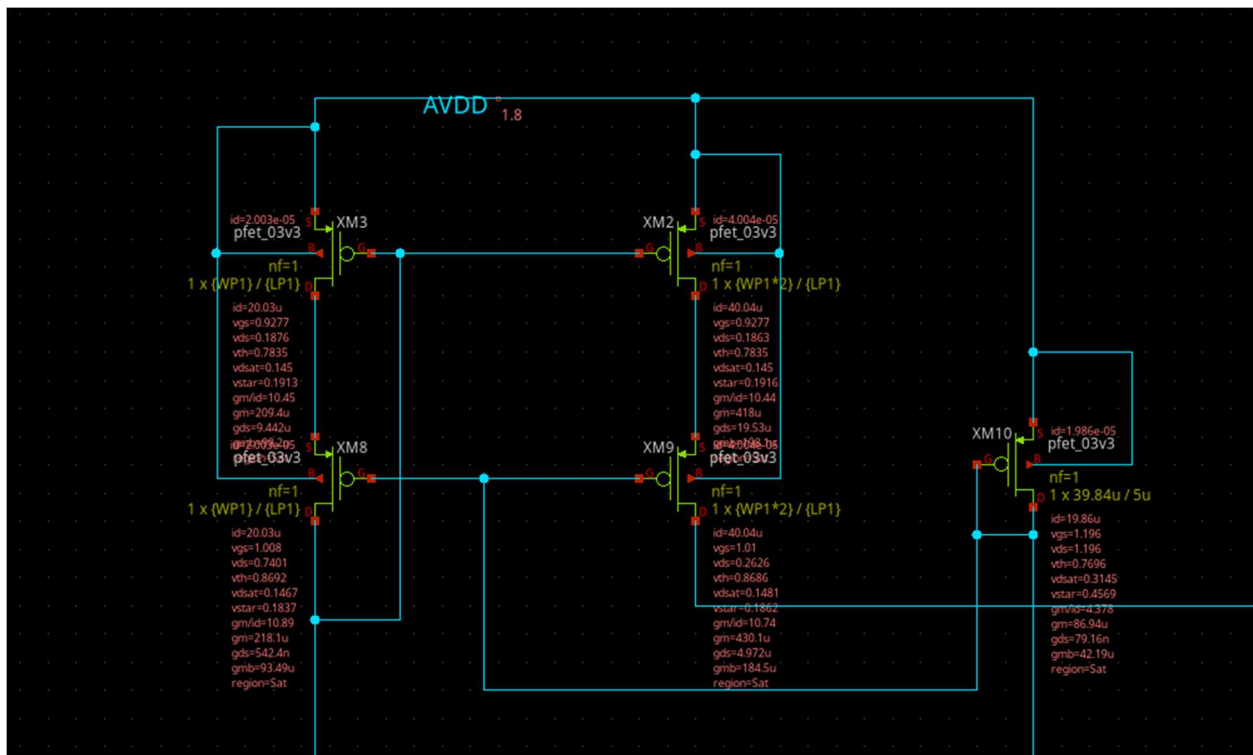
So the magic battery $v_{sg} = 1.2V$

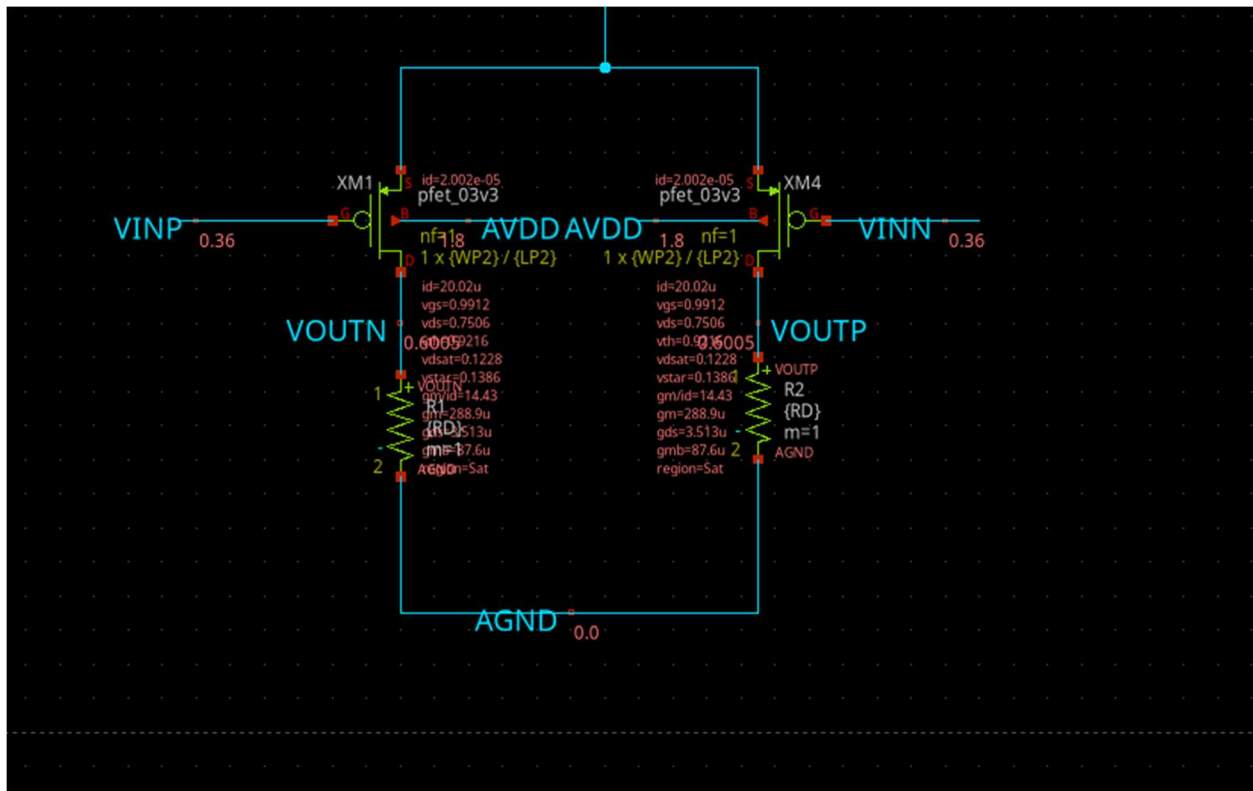
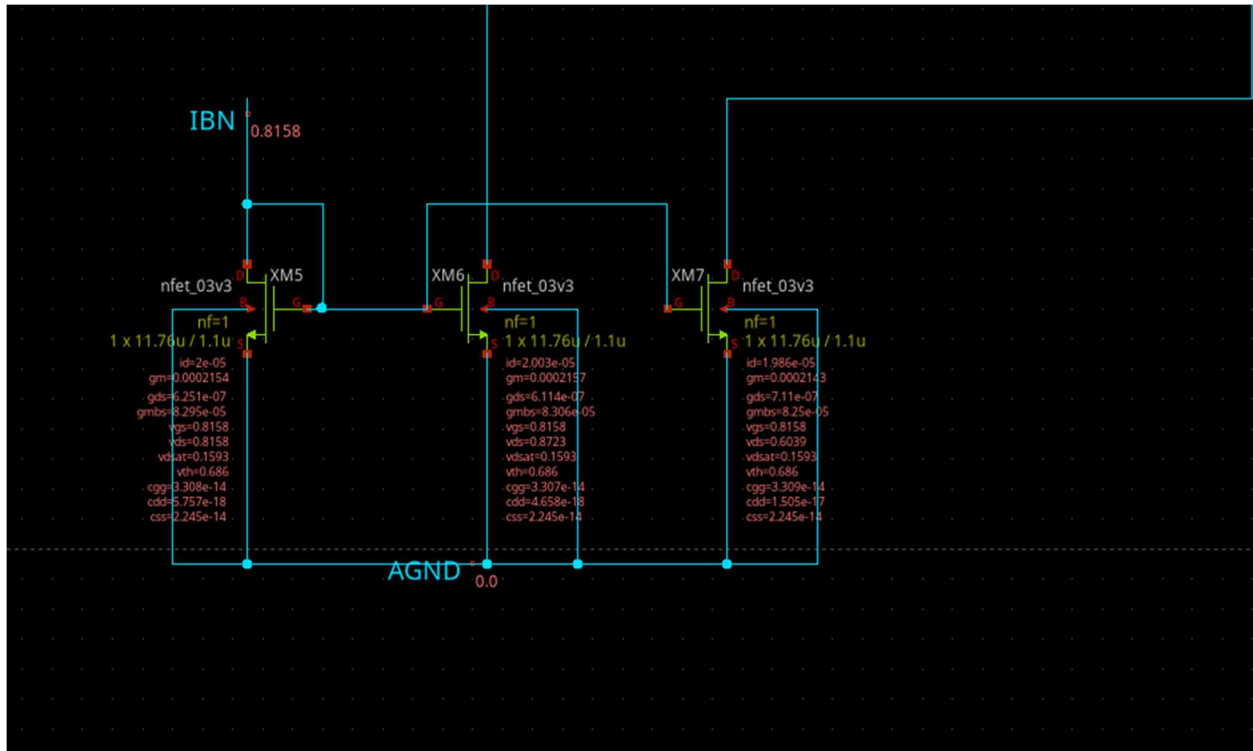


NMOS mirroring devices



We need to copy the current two times using sinking devices here are the sizing for them







Avcm vs frequency

Bode plot

